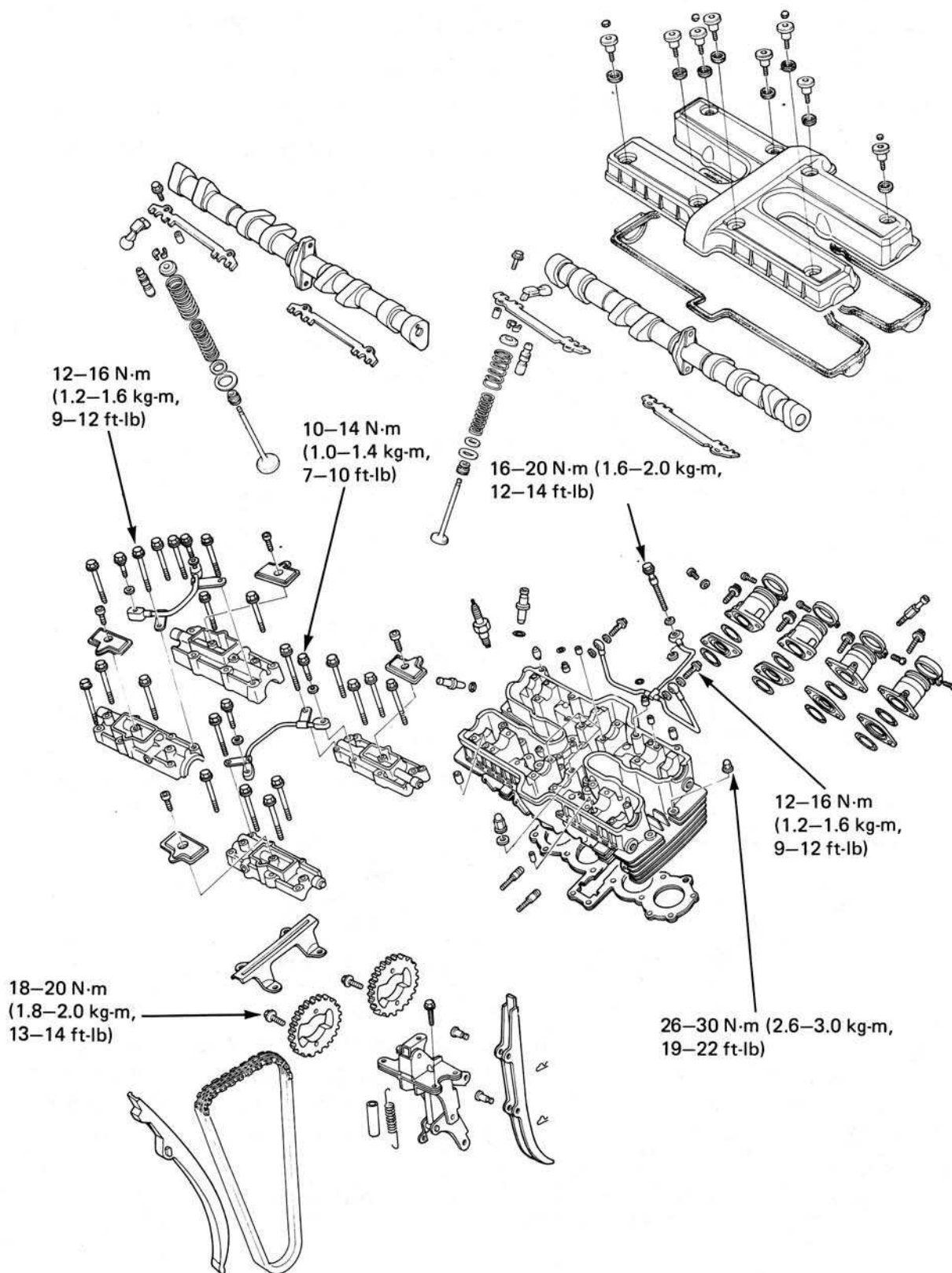




HONDA CBX750F

6. Cylinder Head & Valves





SERVICE INFORMATION	6-1	VALVE GUIDE REPLACEMENT	6-13
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CYLINDER HEAD REMOVAL	6-8	HYDRAULIC TAPPET INSTALLATION	6-18
CYLINDER HEAD DISASSEMBLY	6-11	CAMSHAFT INSTALLATION	6-19

SERVICE INFORMATION

GENERAL

- The engine uses hydraulic valve tappets the eliminate manual valve adjustments
- The hydraulic tappets have de-foaming chamber. Before assembling, fill the chamber with clean engine oil
- Do not turn the camshaft before installing camshaft holders and filling the de-foaming chamber with engine oil, when you adjust the valve timing
- Whenever the camshaft is removed, bleed air from the tappets thoroughly (See page 6-18)
- Lubricate the camshaft journals and cam loves with molybdenum disulfide grease for initial lubrication
- The camshaft holder are identified by the respective marking (IN R: Intake right, IN L: intake left, EX R: Exhaust right, EX L: exhaust left)

SPECIFICATIONS

ITEM			STANDARD	SERVICE LIMIT
Compression pressure			1,200+/- 200 kPa (12.0 +/- 2.0 kg/cm ² , 171+/- 28 psi)	-
Camshaft	Cam height	IN,EX	32.829-33.069 mm (1.2925-1.3019 in)	32.75 mm (1.289 in)
	Oil Clearance	IN1, IN4, EX1 ,EX4	0.020-0.062 mm(0.0008-0.0024 in)	0.09 mm (0.004 in)
		IN2, IN3, EX2, EX4	0.055-0.097 mm (0.0022-0.0038 in)	0.12 mm (0.005 in)
	Runout	IN, EX	-	0.10 mm (0.004 in)
Valve Spring	Free Length	IN, EX Outer	41.7 mm (1.64 in)	40.2 mm (1.58 in)
		IN, EX Inner	36.83 mm (1.45 in)	35.5 mm (1.40 in)
	Preload length	IN, EX Outer	29.5-33.5 kg/24.9 mm (65.0-73.9 lb/ 0.980 in)	-
		IN, EX Inner	12.5-14.5 kg /21.1 mm (27.6-32.0 lb/0.831 in)	-
Valve, Valve guide	Valve Stem OD	IN	4.975-4.990 mm (0.1959-0.1965 in)	4.97mm (0.195 in)
		EX	4.955-4.970 mm (0.1951-0.1957 in)	4.94mm (0.194 in)
	Valve Guide ID	IN, EX	5.0-5.012 mm (0.1969-0.1973 in)	5.04mm (0.198 in)
	Stem-to-guide clearance	IN	0.010-0.037 mm (0.0004-0.015 in)	0.07mm (0.003 in)
		EX	0.030-0.057mm (0.0012-0.0022 in)	0.09 mm (0.004 in)
Cylinder Head	Warpage			0.10 mm (0.004 in)
	Valve seat width	IN, EX	0.9-1.1mm (0.035-0.043 in)	1.5 mm (0.06 in)



HONDA CBX750F

6. Cylinder Head & Valves

TORQUE VALUES

Camshaft holder		12-16 Nm (1.2-1.6 kg-m, 9-12 ft lb)
Cam Sprocket		18-20 Nm (1.8-2.0 kg-m, 13-14 ft lb)
Cylinder Head		26-30 Nm (2.6-3.0 kg-m, 19-22 ft lb)
Spark Plug		12-18 Nm (1.2-1.8 kg-m, 9-13 ft lb)
Oil pipe bolt	7mm	10-14 Nm (1.0-1.4 kg-m, 7-10 ft lb)
	8mm	12-16 Nm (1.2-1.6 kg-m, 9-12 ft lb)
	10mm	23-27 Nm (2.3-2.7 kg-m, 17-20 ft lb)

TOOLS

Special

Valve guide reamer	07984-MA60000
Valve guide drive	07942-MA60000
Hydraulic tappet bleeder	07973-MJ00000

Common

Valve spring compressor	07757-0010000
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Valve seat cutter

Valve seat cutter, 24.5 mm (EX 45°)	07780-0010100
Valve seat cutter 27.5 mm (IN 45°)	07780-0010200
Valve seat cutter 25 mm (EX 32°)	07780-0012000
Valve seat cutter 28 mm (IN 32°)	07780-0012100
Valve seat cutter 22 mm (EX 60°)	07780-0014202
Valve seat cutter 26 mm (IN 60°)	07780-0014500
Valve seat cutter holder, 5 mm	07781-0010400

TROUBLESHOOTING

Engine top end problems are usually performance related and can be diagnosed by a compression test, or are engine noises which can be traced to the top end with a sound rod or stethoscope.

Low compression or uneven compression

1. Valves

- faulty hydraulic tappet
- burned or bent valves
- incorrect valve timing
- broken valve spring

2. Cylinder head

- Leaking or damaged head gasket
- Warped or cracked cylinder head

3. Cylinder and Piston (refer to Section 7)

Compression too high

- Excessive carbon building up on piston head or combustions chamber

Excessive noises

1. Faulty hydraulic valve tappet system

- Low engine oil level
- Contaminated oil
- Low oil pressure
- Damaged hydraulic tappet

2. Sticking valve or broken valve spring

3. Damaged or worn camshaft

4. Loose or worn cam chain

5. Worn or damaged cam chain tensioner

6. Worn cam sprocket teeth



6. Cylinder Head & Valves

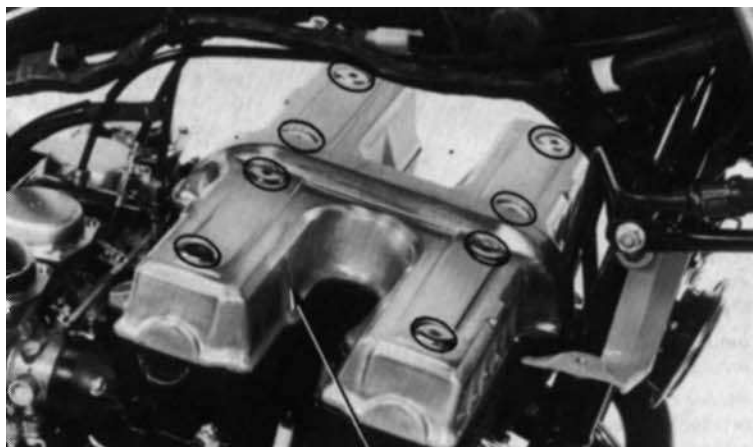
CAMSHAFT REMOVAL

Remove the following parts

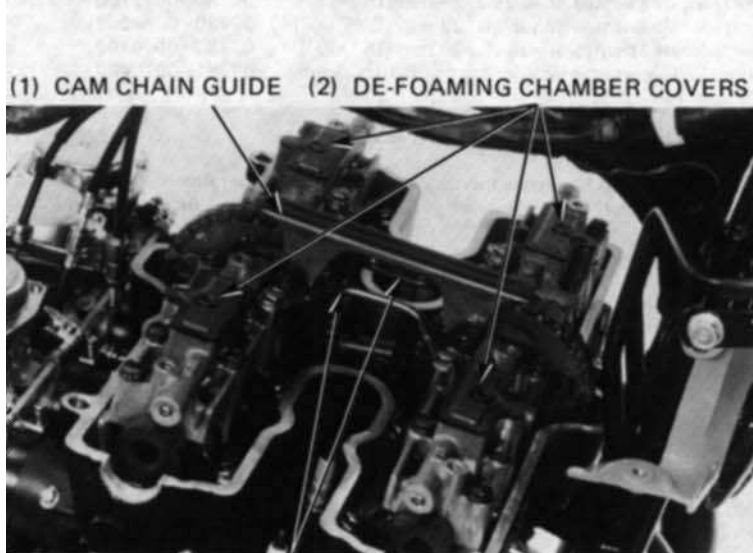
- faring (page 13-3)
- fuel tank (page 4-15)
- ignition coil

Remove the cam chain guide and oil pipes.
Remove the de-foaming chamber covers.

Loosen the cam chain by pushing the cam chain tensioner lock pins down, and pulling the lock plate up until the lock plate rests on the lock pin shoulder as shown.

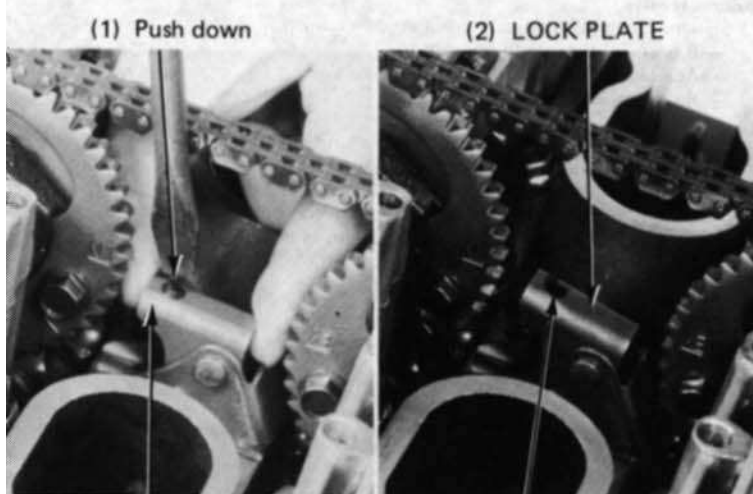


(1) CYLINDER HEAD COVER



(1) CAM CHAIN GUIDE (2) DE-FOAMING CHAMBER COVERS

(3) OIL PIPES



(1) Push down

(2) LOCK PLATE

(3) Pull up

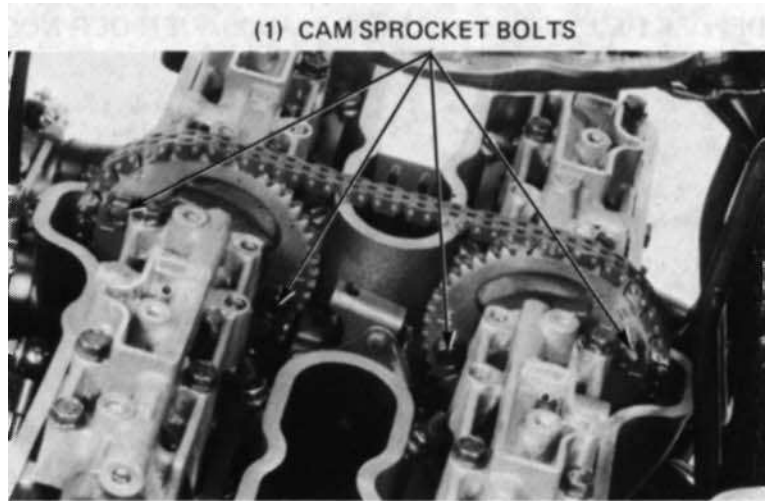
(4) LOCK PIN



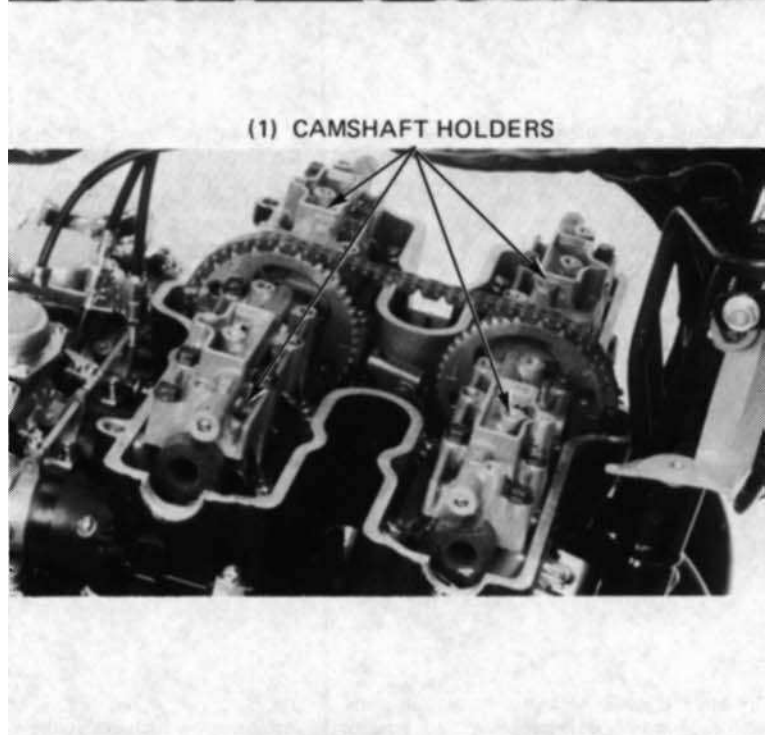
HONDA CBX750F

6. Cylinder Head & Valves

Remove the right crankshaft cover.
Remove the two cam sprocket bolt bolts first from both sprockets, turn the crankshaft clockwise, then remove the other two sprocket bolts.

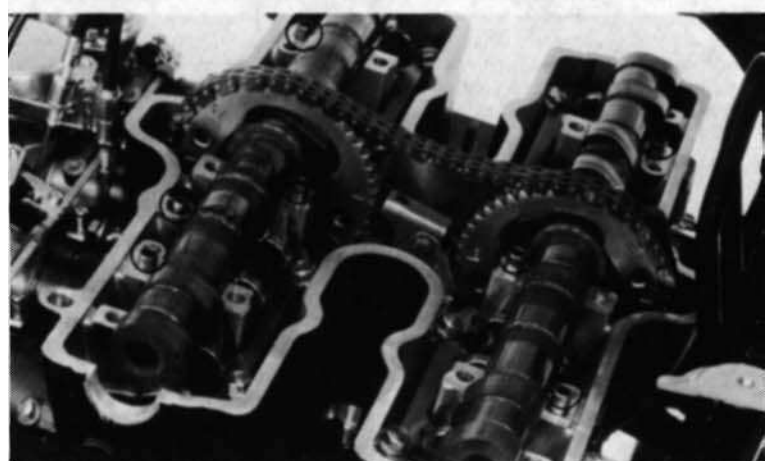


Slide the cam sprockets off the camshaft sprocket flanges.
Remove the camshaft holder bolts, and the holder.



NOTE:
Loosen the holder bolts in 2-3 steps in criss-cross pattern.

Remove the dowel pins and O-Rings.
Remove the camshafts and cam sprockets.





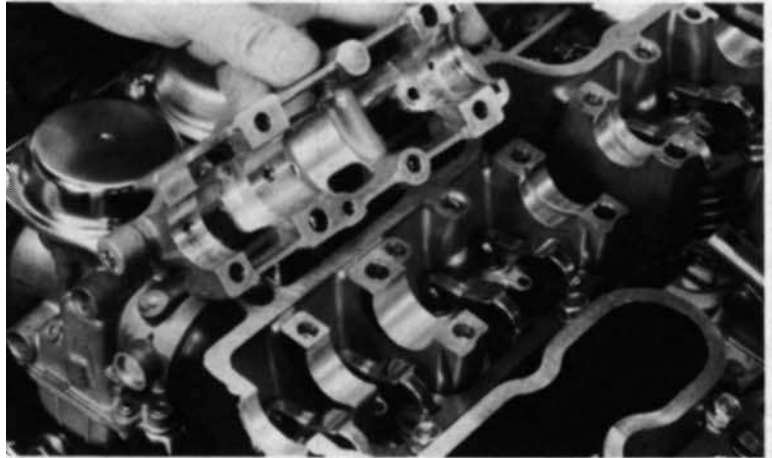
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6. Cylinder Head & Valves

INSPECTION

CAMSHAFT BEARING SURFACE

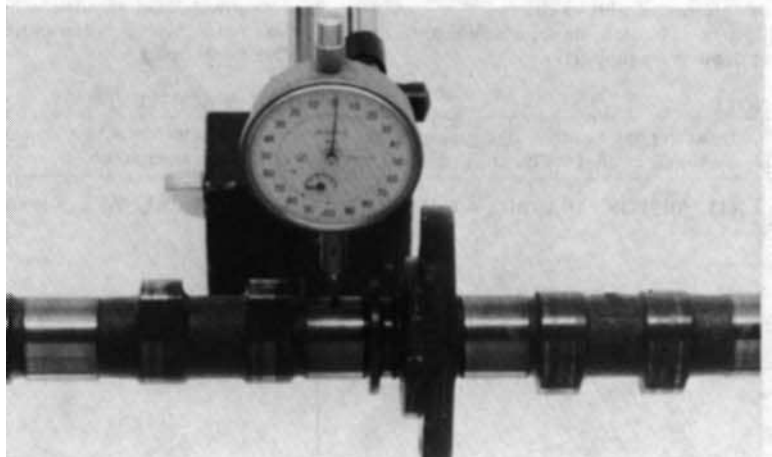
Inspect the cam bearing surfaces for scoring, scratches, or evidence of insufficient lubrication. Also inspect the bearing surfaces of the camshaft holders.



CAMSHAFT RUNOUT

Check the camshaft run-out with a dial indicator. Support both end of the camshaft with V blocks.

SERVICE LIMIT: 0.10 mm (0.004 in)



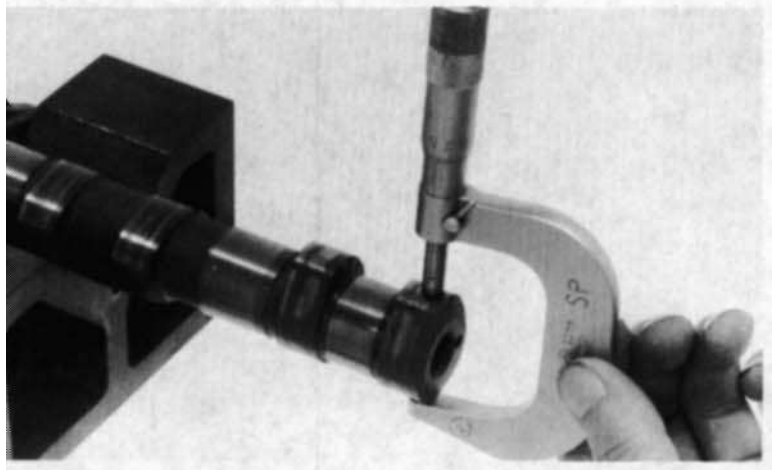
CAM LOBE HEIGHT

Check the camshaft lobes for wear or damage. If the lobes are scored, inspect the rocker arm surfaces also.

Measure the cam lobe height with a micrometer

SERVICE LIMIT:

IN, EX: 32.75 mm (1.289 in)





HONDA CBX750F

6. Cylinder Head & Valves

CAMSHAFT OIL CLEARANCE

Wipe any oil from the camshaft journals.

Lay a strip of plastigauge lengthwise on top of each camshaft journal.

Install the camshaft holders and tighten in a criss-cross pattern

NOTE:

Do not rotate the camshaft when using plastigauge.

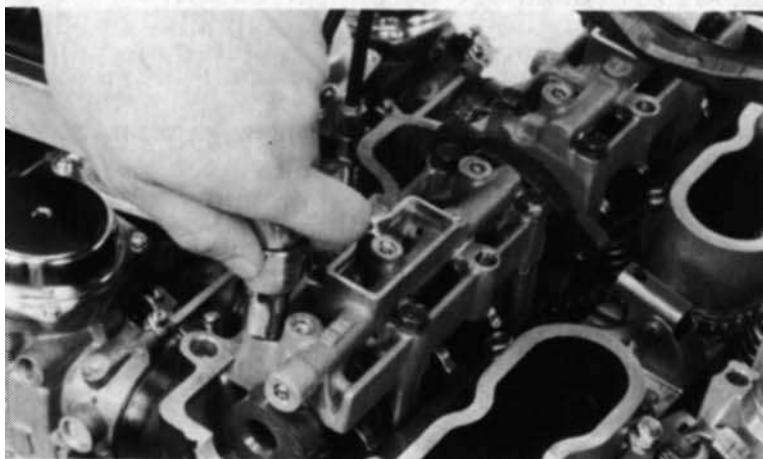
TORQUE: 12-16 Nm (1.2-1.6 kgm, 9-12 ft.lb)

Remove the camshaft holders and measure the width of each plastigauge. The widest thickness determines the oil clearance.

SERVICE LIMITS:

IN1, IN4, EX1, EX4L	0.09 mm (0.004 in)
IN2, IN3, EX2, EX3	0.12 mm (0.005 in)

When the service limits are exceeded, replace the camshaft and recheck the oil clearance. Replace the cylinder head and camshaft holders if the clearance still exceeds the service limits.





6. Cylinder Head & Valves

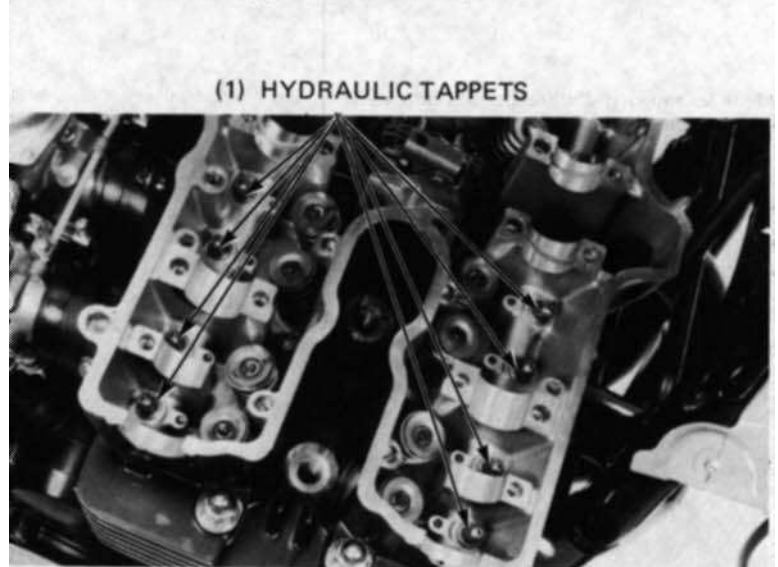
HYDRAULIC TAPPET REMOVAL

Remove the camshafts (page 6-3)
Remove the rocker arms.
Remove the rocker arm holder bolts and the holders.
Remove the dowel pins.

Remove the hydraulic tappets.



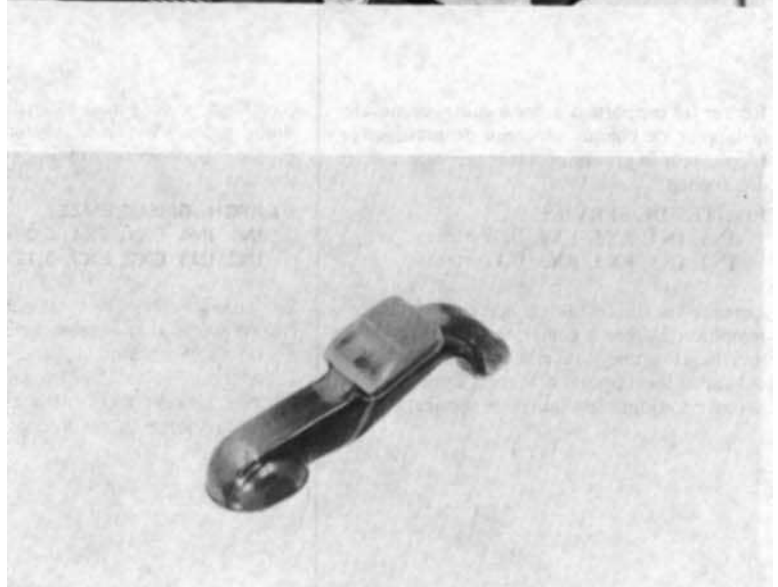
(2) ROCKER ARMS



INSPECTION

ROCKER ARM

Inspect the rocker arm followers for damage or abnormal wear, and replace if necessary.





HONDA CBX750F

6. Cylinder Head & Valves

HYDRAULIC TAPPET

Inspect the hydraulic tappet for wear or damage or for clogged oil hole.

CAUTION:

- **Never attempt to disassemble the tappets**
- **Always use the special tool when bleeding the tappets. Use of wire can cause damage to them.**

Measure the free length of each hydraulic tappet as follows:

Attach the Hydraulic Tappet Bleeder to the hydraulic tappet and compress and extend the hydraulic tappet slowly in a jar filled with kerosene.

NOTE:

Hold the hydraulic tappet upright while compressing and extending the hydraulic tappet.

Continue operated the hydraulic tappet until there are no air bubbles from the hydraulic tappet and it does not make any further action. Remove the hydraulic tappet and try to compress quickly the tappet by hand. Measure the compression stroke with the dial gauge on the flat place.

COMPRESSION STROKE: 0-0.2 mm

NOTE:

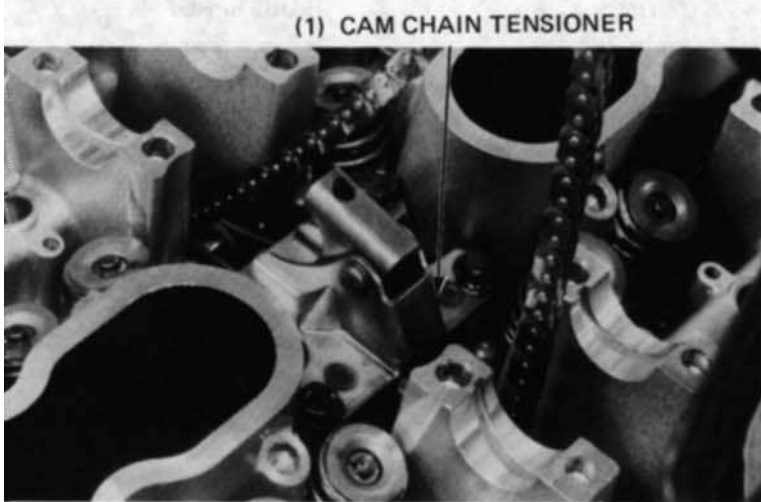
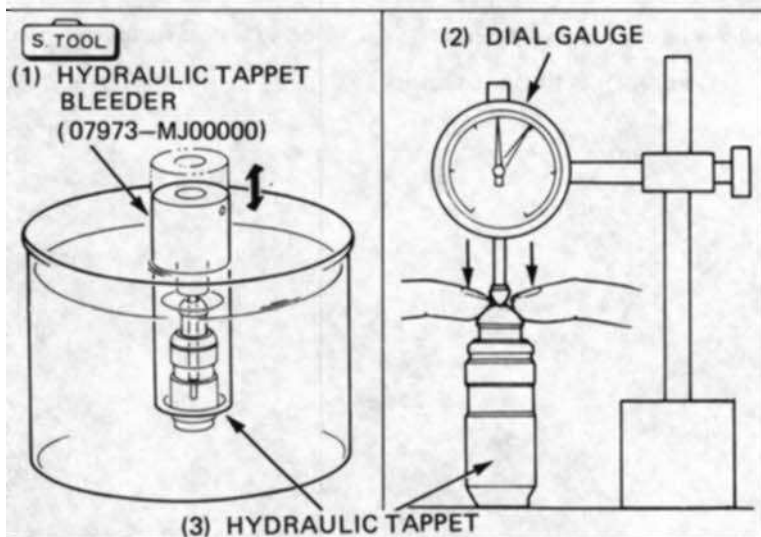
Keep the hydraulic tappet below the surface of kerosene while priming the hydraulic tappet.

CYLINDER HEAD REMOVAL

Remove the following parts:

- carburettor (section 4)
- exhaust system (section 5)
- hydraulic tappets (page 6-7)

Remove the four cam chain tensioner mounting bolts.





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6. Cylinder Head & Valves

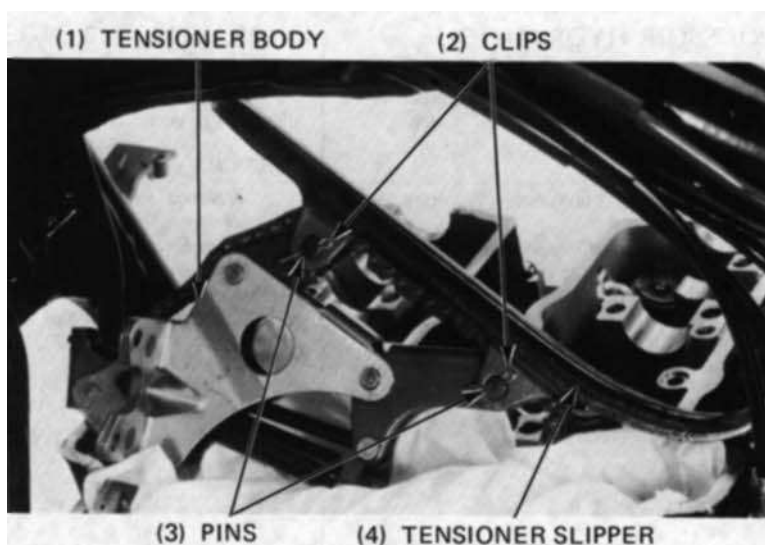
Remove the clips and pins from the tensioner.

NOTE:

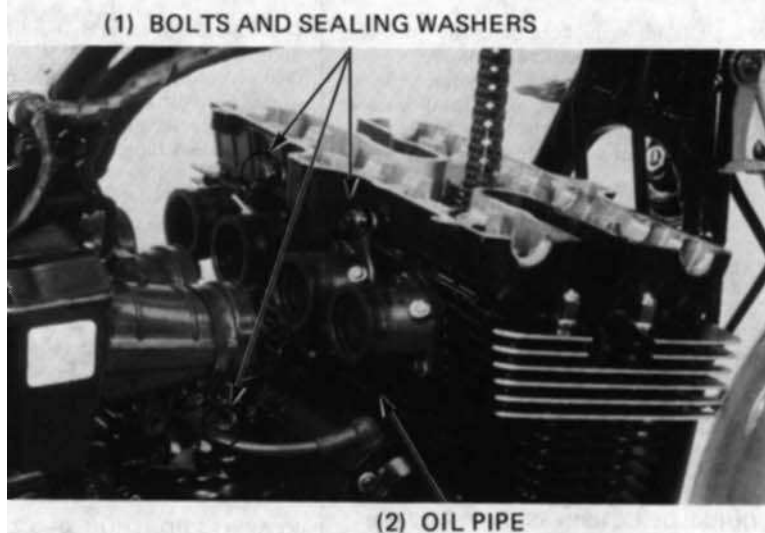
Use care when removing the clips and pins to prevent them from falling into the crankcase.

Place a piece of wire through the cam chain. Tie it so the chain does not fall into the crankcase.

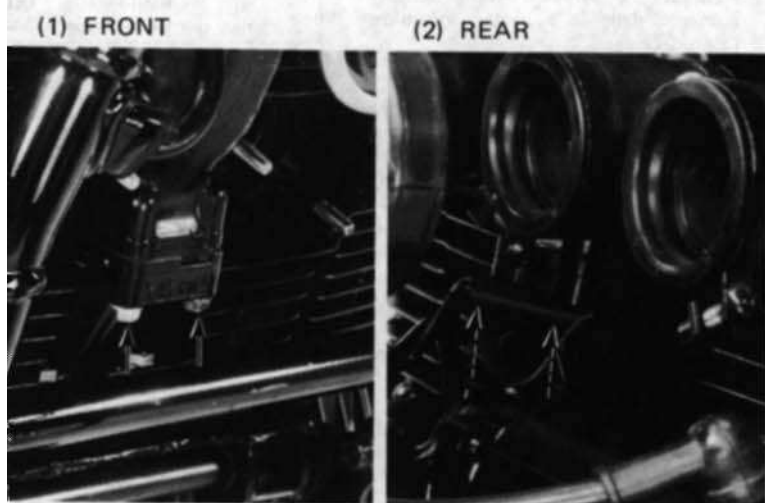
Separate the tensioner body and slipper and remove them.



Remove the oil pipe bolts and sealing washers.



Remove the front and rear cylinder head mount bolts.
Remove the oil pipe.





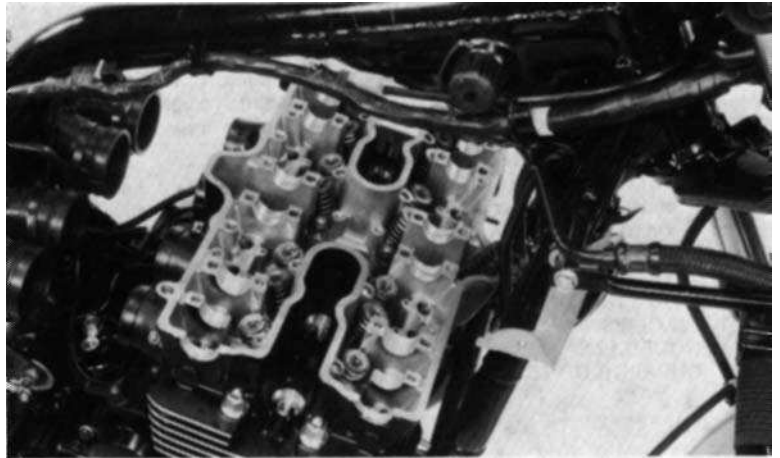
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6. Cylinder Head & Valves

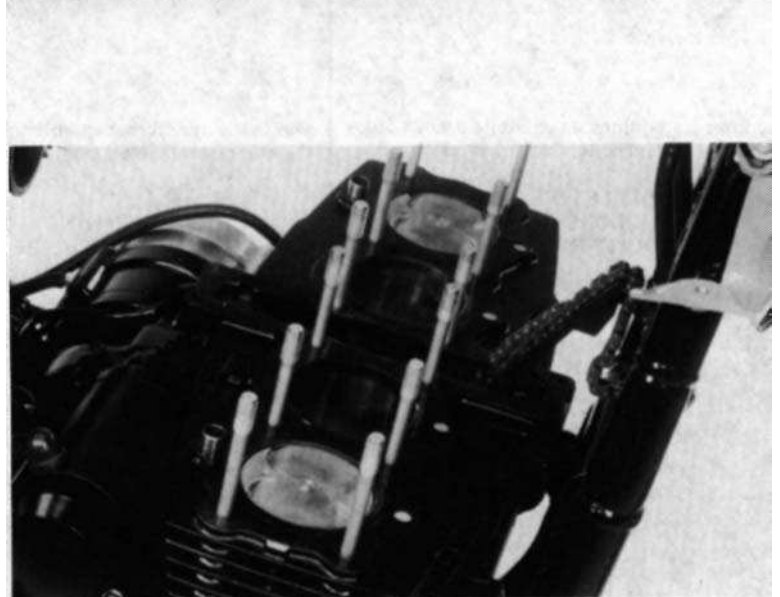
Remove the cylinder head cap nuts and washer.

NOTE:

Loosen the nut in 2-3 steps in a crisscross pattern to prevent cylinder head warpage.

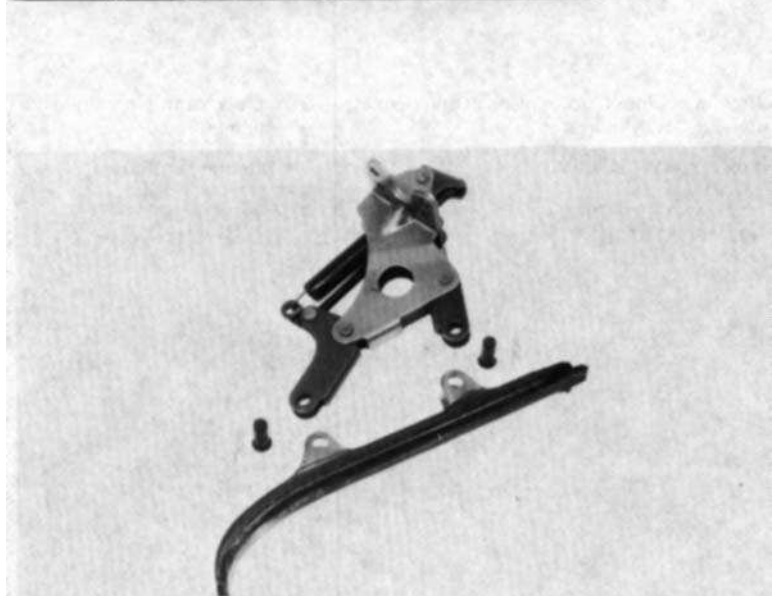


Remove the cylinder head
Remove the cylinder head gasket, dowel pins, and cam chain guide.



CAM CHAIN GUIDE AND TENSIONER INSPECTION

Inspect the cam chain guide and tensioner for damage or excessive wear.
Inspect the cam chain tensioner slipper for damage or excessive wear.
Replace the tensioner body if the spring is weak or has been damaged.





CYLINDER HEAD DISASSEMBLY

Remove the valve cotters, retainers, springs and valves using a valve spring compressor.

CAUTION;

- *To prevent the loss of tension do not compress the valve springs more than necessary to remove the cotters.*
- *Remove valve spring compressor large spring retainer before using to avoid damaging the cylinder head.*

NOTE;

Mark all disassembled parts to ensure correct reassembly.

Remove the carbon deposits from the combustion chamber.

Clean the head gasket surfaces thoroughly

CAUTION:

Avoid damaging the gasket surfaces.

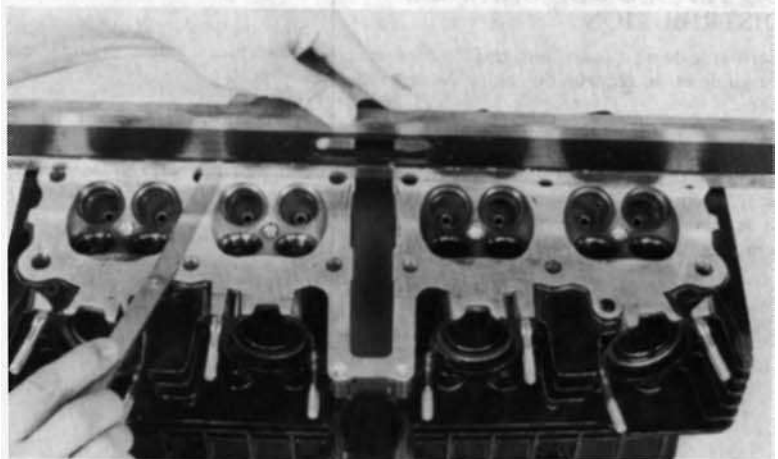
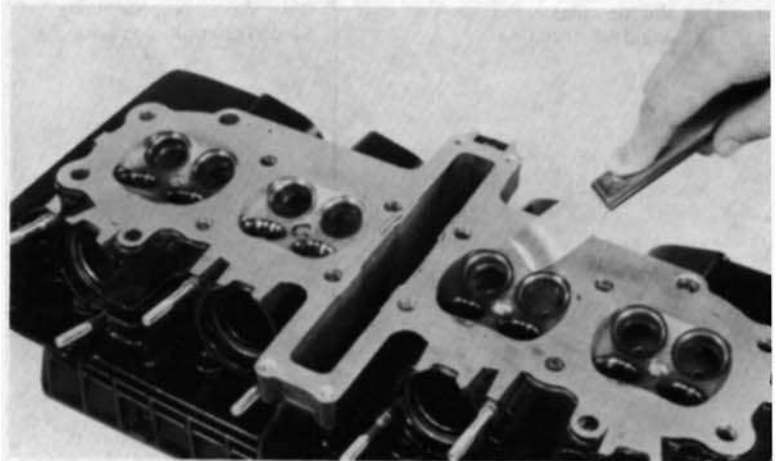
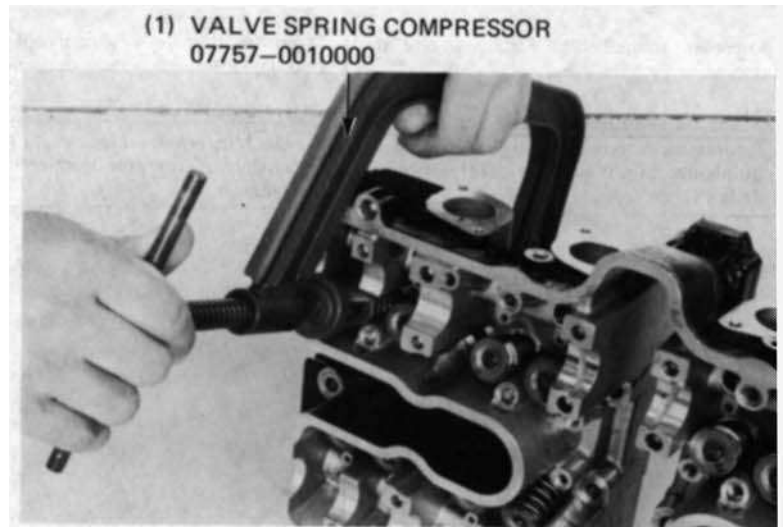
INSPECTION

CYLINDER HEAD

Check the spark plug holes and valve areas for cracks.

Check the cylinder head for warpage with a straight edge and a feeler gauge in an X pattern.

SERVICE LIMIT: 0.10mm (0.004 in)





HONDA CBX750F

6. Cylinder Head & Valves

VALVE SPRING FREE LENGTH

Measure the free length of the inner and outer valve springs.

SERVICE LIMITS

INNER SPRING:35.5 mm (1.40 in)

OUTER SPRING:40.2 mm (1.58 in)

Replace them if they are shorter than the service limit.

VALVE STEM TO GUIDE CLEARANCE

Inspect each valve for bending, burning, scratches or abnormal stem wear.
Check the valve movement in the guide.
Measure and record each valve stem O.D.

SERVICE LIMITS

IN: 4.97 mm (0.195 in)

EX: 4.94 mm (0.194 in)

NOTE:

Ream the guides to remove any carbon build up before checking the clearance.

Measure and recorded each valve guide using a ball gauge or inside micrometer.

SERVICE LIMITS:

IN: 5.04 mm (0.198 in)

EX: .5.04 mm (0.198 in)

Subtract each valve stem O.D. from the corresponding guide I.D. to obtain the stem to guide clearance.

SERVICE LIMITS

IN: 0.07 mm (0.003 in)

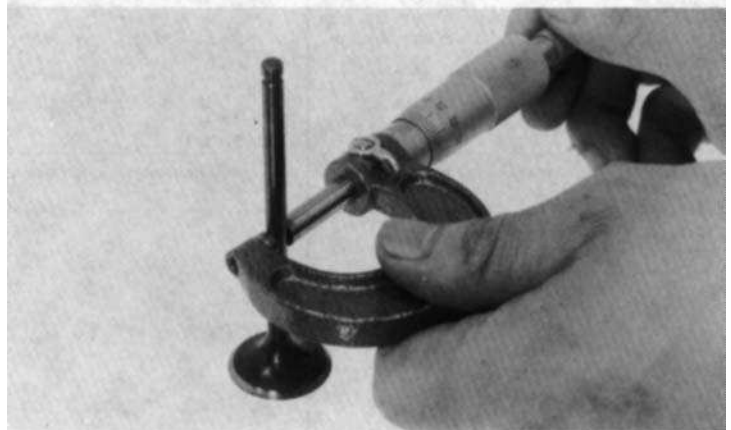
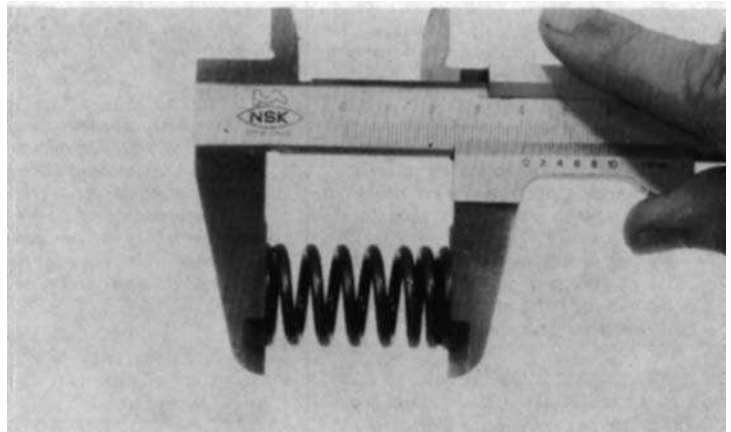
EX: 0.09 mm (0.004 in)

If the stem to guide clearance exceeds the service limits, determine if a new guide with standard dimensions would bring the clearance within tolerance. If so, replace any guides as necessary and ream to fit.

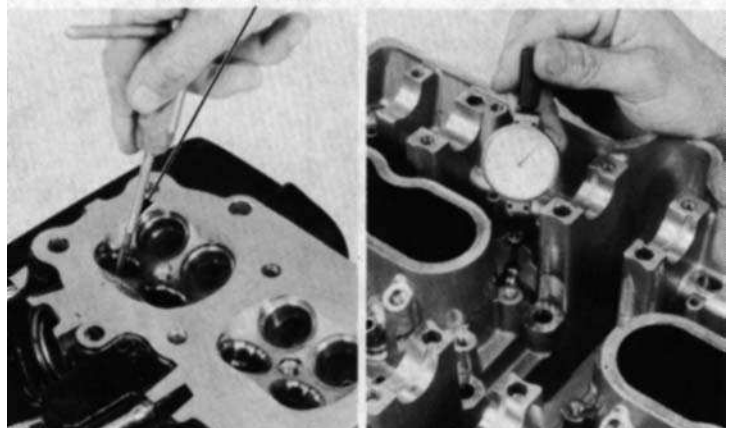
If the stem to guide clearance exceeds the service limits with new guides, replace the valves and the guides.

NOTE:

Reface the valve seat whenever the valves guides are replaced (page 6-14)



(1) VALVE GUIDE REAMER
07984-MA60000





VALVE GUIDE REPLACEMENT

Support the cylinder head and drive the guide from the valve port out.

NOTE:

When driving out the valve guide, do not damage the head.

Install an oversize valve guide from the top of the head.

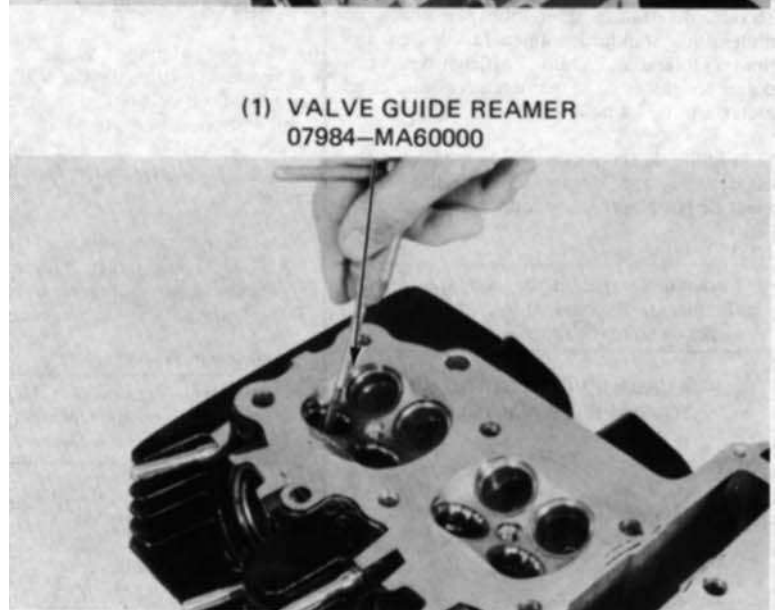
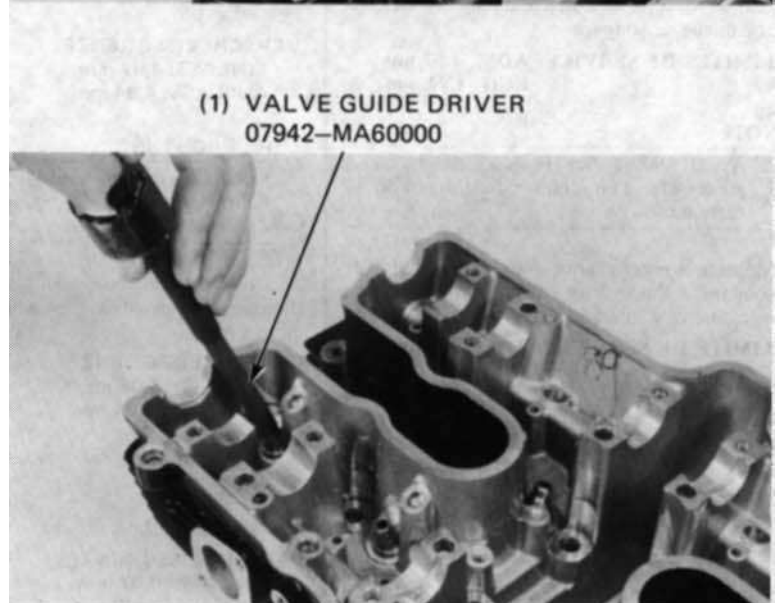
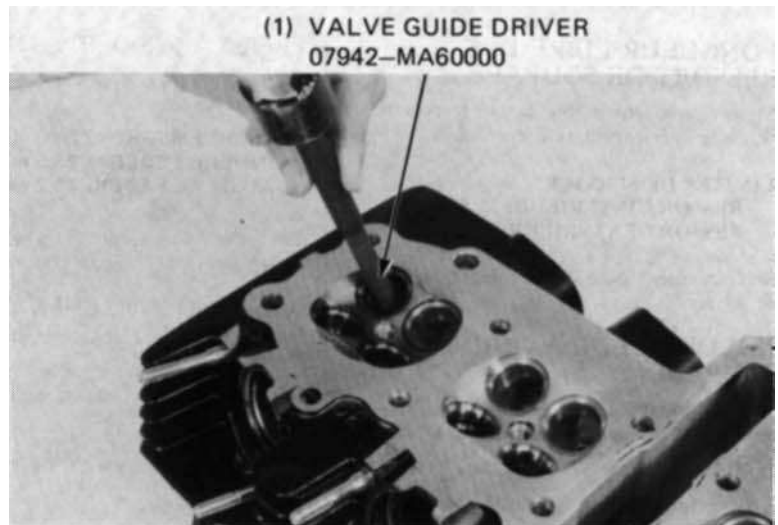
Ream the new valve guide after installation.

NOTE:

- Use cutting oil on the reamer during this operation.
- Always rotate the reamer in the same direction when inserting and removing it.

Reface the valve seat (page 6-14)

Clean the cylinder head thoroughly to remove any metal particles.





VALVE SEAT INSPECTION/ REFACING

Clean all intake and exhaust valves thoroughly to remove carbon deposits.

Apply a light coating of valve lapping compound to each valve face. Lap each valve and seat a few times with light pressure using a rubber hose or other hand lapping tool.

NOTE:

Take care not to allow the compound to enter between the valve stem and guide. After lapping, wash out the compound completely and apply a coat of engine oil to the valve face and seat.

Remove the valve and inspect the face.

CAUTION:

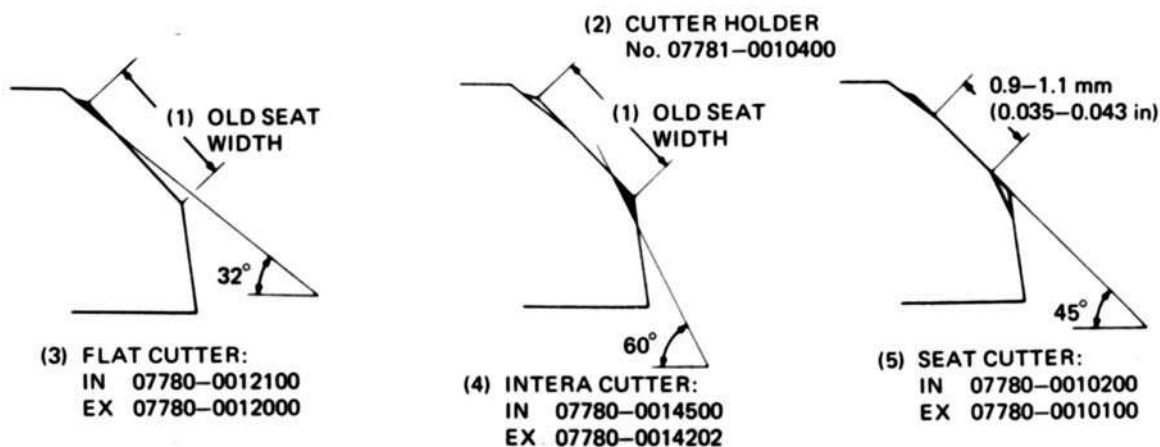
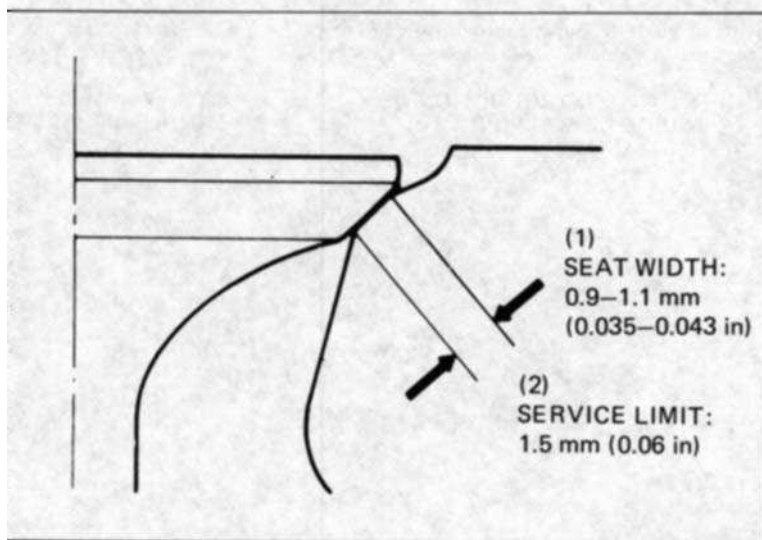
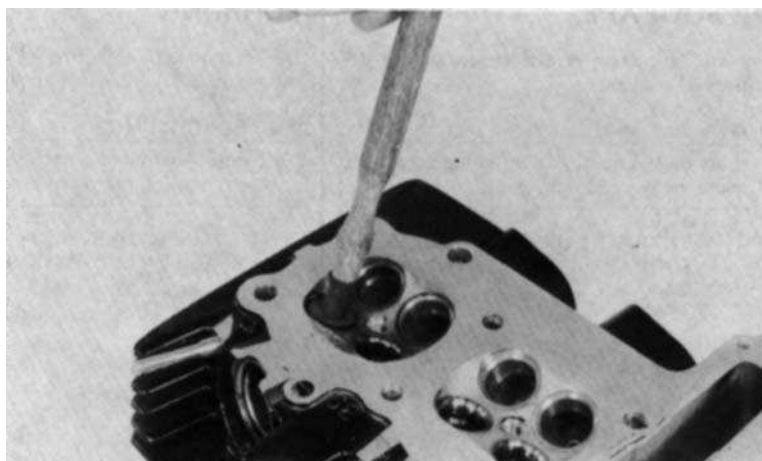
The valves cannot be ground, if the valve face is rough, worn unevenly, or contacts the seat improperly, the valve must be replaced.

Inspect the valve seat. If the seat is too wide, too narrow, or has low spots, the seat must be ground.

NOTE:

Follow the refacer manufacturers operating instructions.

After cutting the seat, apply lapping compound to valve face, and lap the valve using light pressure. After lapping, wash any residual compound off the cylinder head and valve.





6. Cylinder Head & Valves

CYLINDER HEAD ASSEMBLY

Install new valve stem seals.

Lubricate each valve stem with molybdenum disulfide grease and insert the valve into the valve guide.

NOTE:

To avoid the damage to the stem seal, turn the valve slowly when inserting.

Install the valve springs and retainers.

NOTE:

Install the valve spring with the tightly wound coils facing the cylinder head.

Install the valve cotters.

CAUTION:

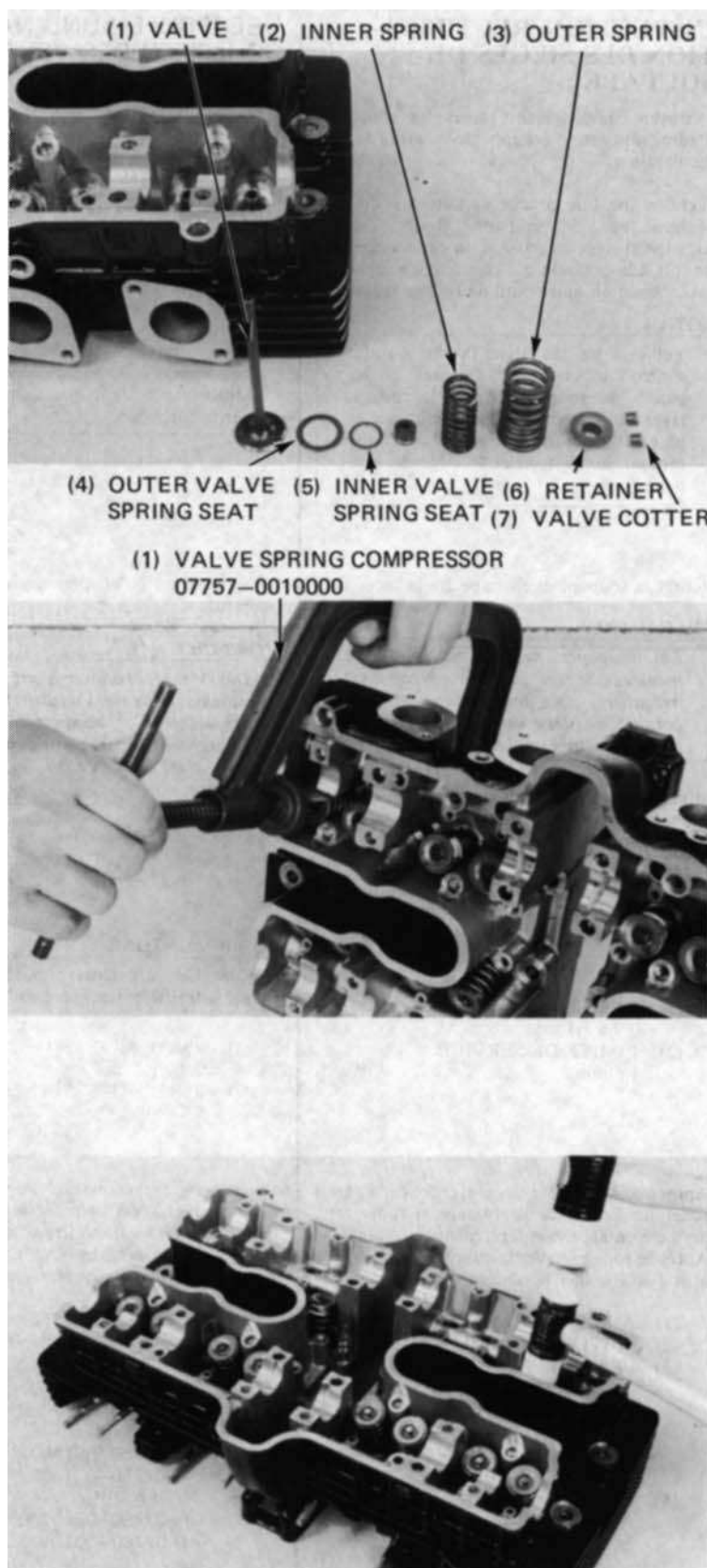
To prevent loss of tension, do not compress the valve spring more than necessary to install the valve cotters.

Tap the valve stems gently with a soft hammer to firmly seat the cotters.

NOTE:

Support the cylinder head above the work bench surface to prevent possible valve damage.

Clean the cylinder head assembly with solvent, after reassembling, then blow through all oil passages with compressed air.





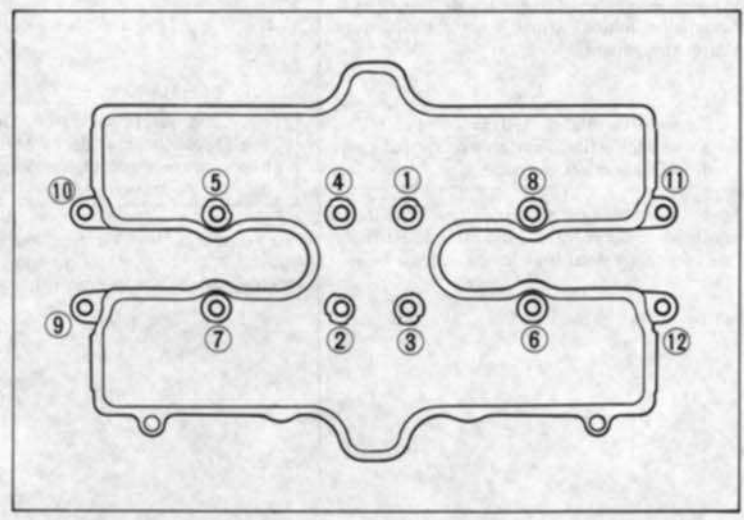
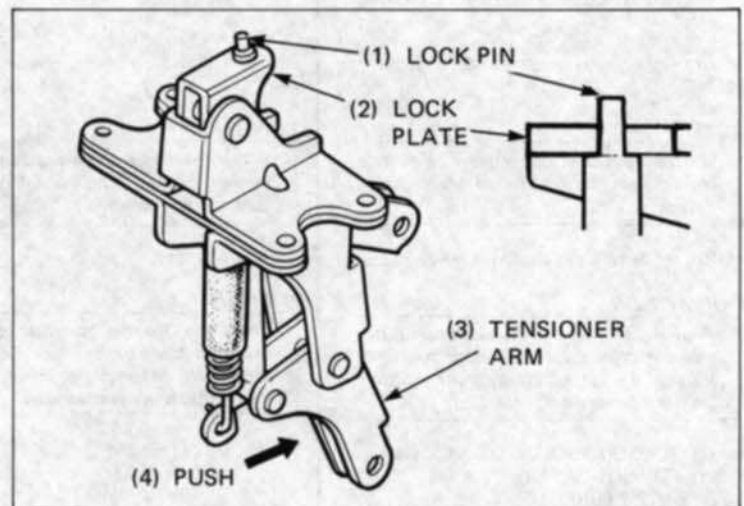
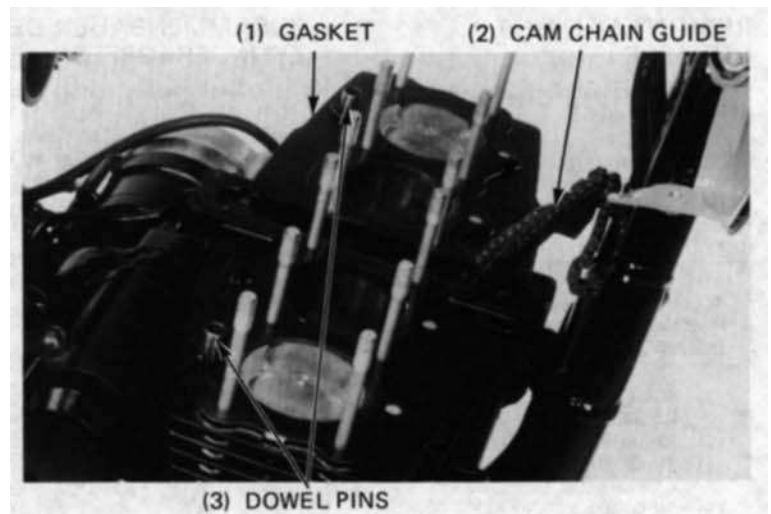
CYLINDER HEAD INSTALLATION

Clean the cylinder surfaces.
Install a new gasket and the dowel pins.
Install the cam chain guide.

Push the tensioner arm and lock the arm by setting the lock pin to the lock plate as shown.

Pull the cam chain guide slightly and push it rearward, then lower the cylinder head.
Set the cam chain guide properly and set the cylinder head.
Apply engine oil to the threads and flanges of the cylinder head cap nuts.
Tighten the cylinder head cap nuts in the sequence shown.

**TORQUES: 26-30Nm
(2.6-3.0 kg-m, 19-22 ft lb)**





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6. Cylinder Head & Valves

Blow the oil pipe with compressed air and install it in position.

Tighten the front and rear cylinder head mounting bolts.

Blow the oil bolts with compressed air. Make sure the sealing washers are in good condition.

Connect the oil pipe with the oil bolts and sealing washers and tighten the bolts.

TORQUE;

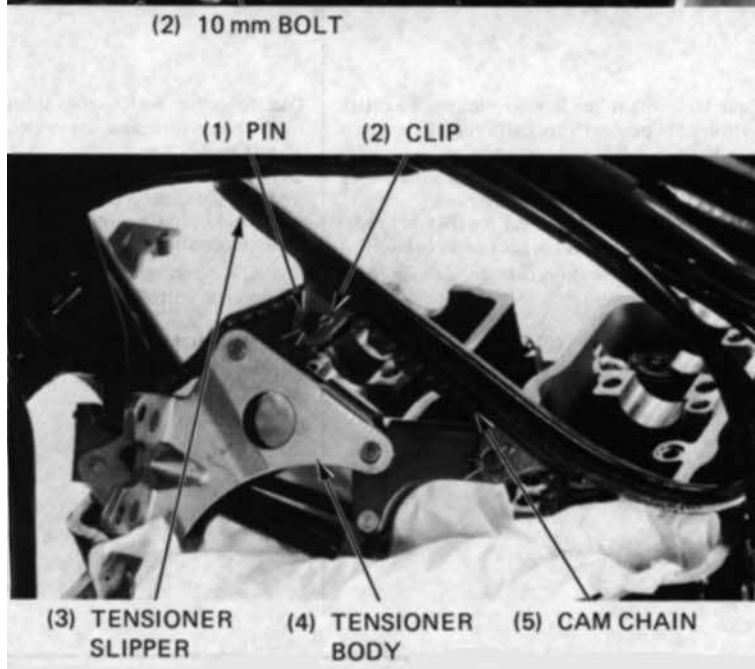
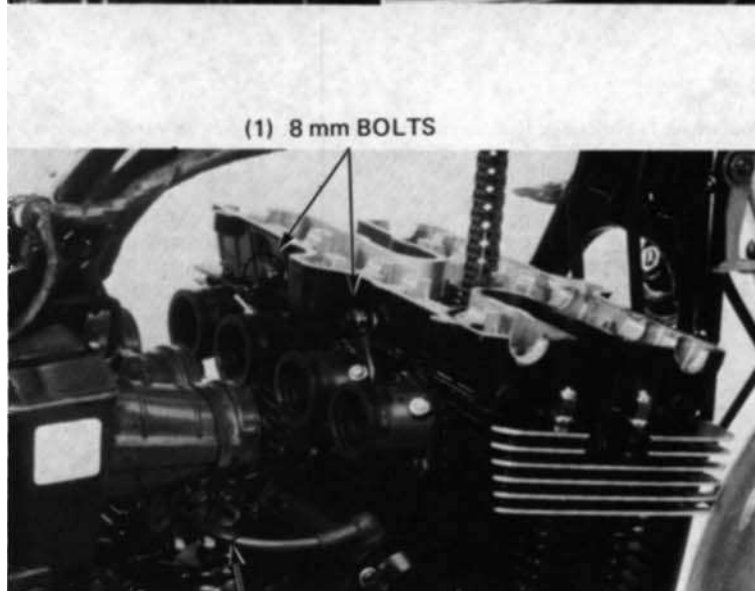
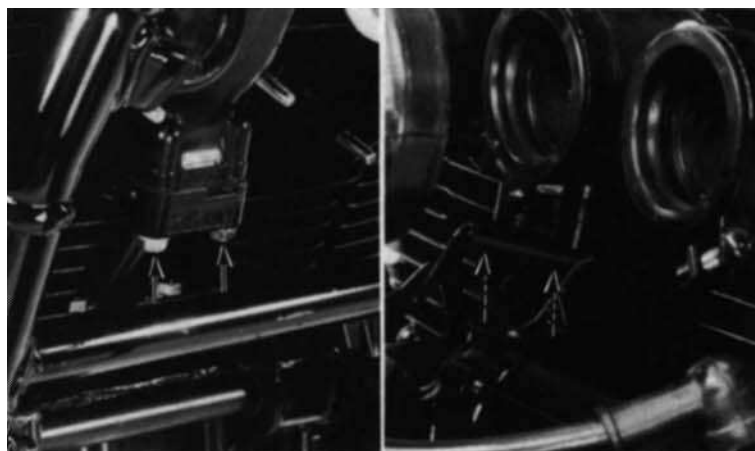
8mm bolts: 12-16Nm
(1.2-1.6 kg-m, 9-12 ft lb)

10mm bolt: 23-27 Nm
(2.3-2.7 kg-m, 17-20 ft lb)

Put the cam chain over the tensioner body and install the tensioner slipper with the clips and pins as shown.

NOTE:

Be careful not to drop the pins and clips into the crankcase.





HONDA CBX750F

6. Cylinder Head & Valves

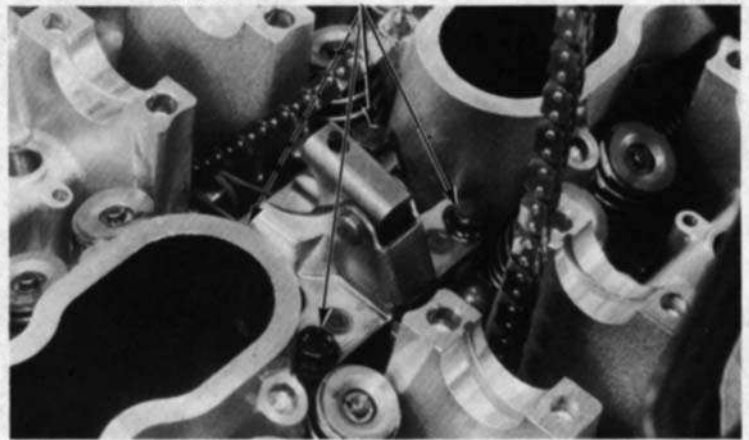
Install and tighten the cam chain tensioner mounting bolts.
Install the carburetors and exhaust system.

HYDRAULIC TAPPET INSTALLATION

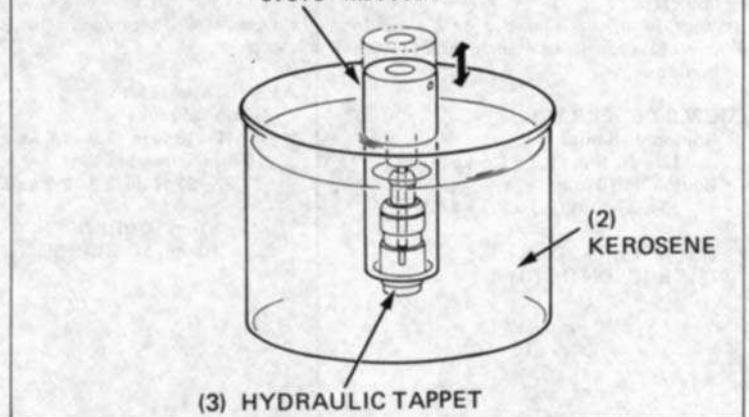
Place the tappet in a jar filled with kerosene. Place the tapped bleed into the tappet. Hold the tappet upright and pump the tappet until air bubbles stop coming out. Remove the tool and try to quickly compress the tappet by hand. You should not be able to compress it more than 0.2mm (0.008 in). Remove the tappet from the fluid keeping it upright.

Fill the all tappet hole up with clean engine oil.
Install the bled hydraulic tappets as described above procedure.
Install the dowel pins into the cylinder head.

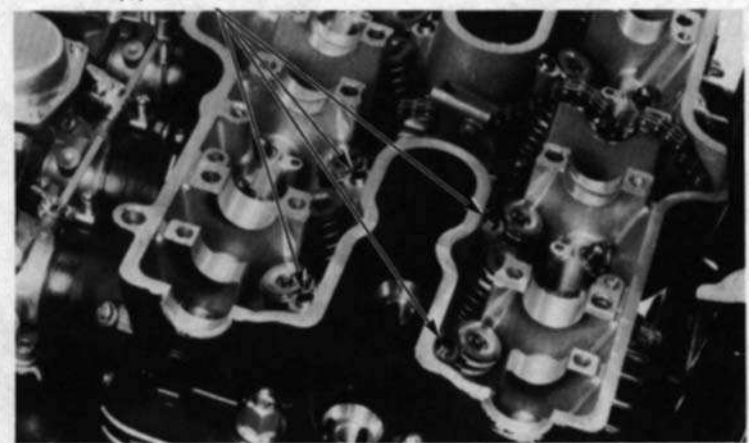
(1) CAM CHAIN TENSIONER MOUNTING BOLTS



S TOOL (1) HYDRAULIC TAPPET BLEEDER 07973-MJ00000



(1) DOWEL PINS



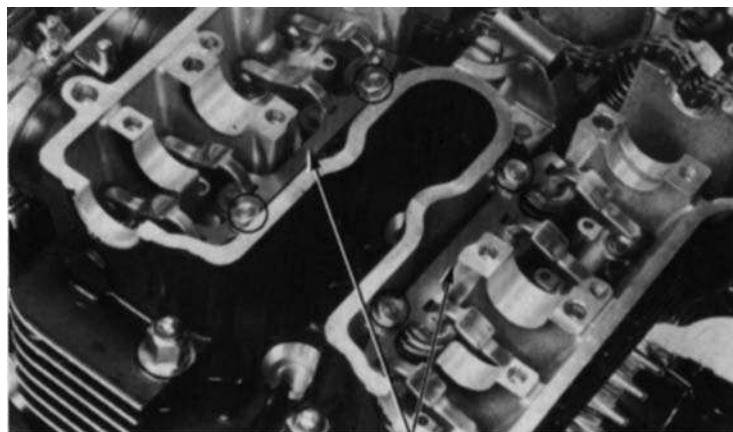


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6. Cylinder Head & Valves

Install the rocker arm holders and tighten the bolts.

Install the rocker arms.



(1) ROCKER ARM HOLDERS

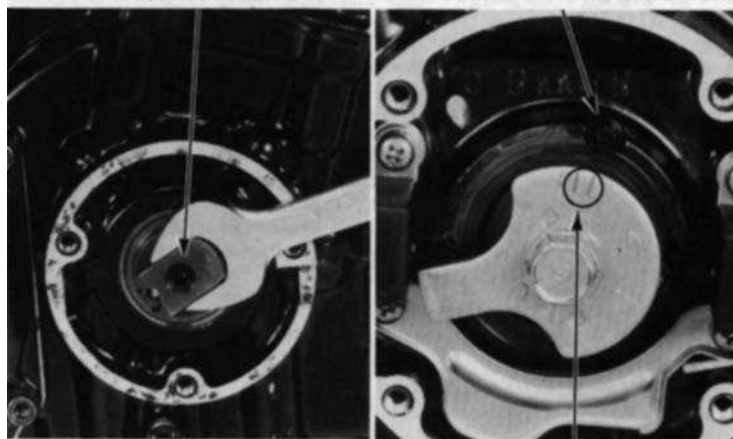
CAMSHAFT INSTALLATION

Remove the pulse generator cover.

Turn the crankshaft clockwise (viewed from the right side) and align the "T" mark on the pulse rotor with the index mark on the crankcase.

(1) CRANKSHAFT
(RIGHT SIDE)

(2) INDEX SIDE



(3) "T" MARK

Lubricate the cylinder head camshaft bearing surfaces with molybdenum disulfide grease.

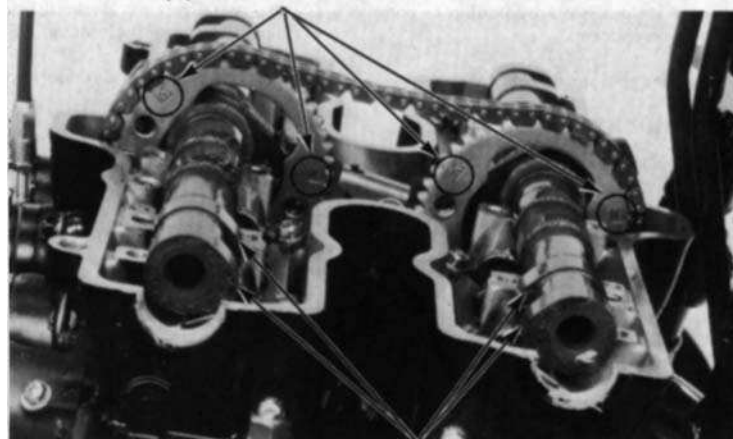
Install the intake and exhaust camshafts and sprockets through the cam chains, so that the timing marks on the sprockets face the right side as shown.

NOTE:

The intake camshaft and "IN" mark and the exhaust camshaft has "EX" mark.

Rotate the camshaft so the No. 4 cylinder cam lobes face each other.

(1) TIMING MARKS



(2) NO. 4 CYLINDER CAM LOBES



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6. Cylinder Head & Valves

Install the two O-rings and dowel pins into the oil passage holes.

Install the eight dowel pins into the camshaft holder bolt holes.

Install each camshaft holder on its original location.

NOTE:

The marks on the camshaft holders mean:

IN R: Intake right

IN L: Intake left

EX R: Exhaust right

EX L: Exhaust left

Temporarily tighten the camshaft holder bolts.

Align the "IN" lines on the intake cam sprocket and the "EX" lines on the exhaust cam sprocket with the top of the cylinder head.

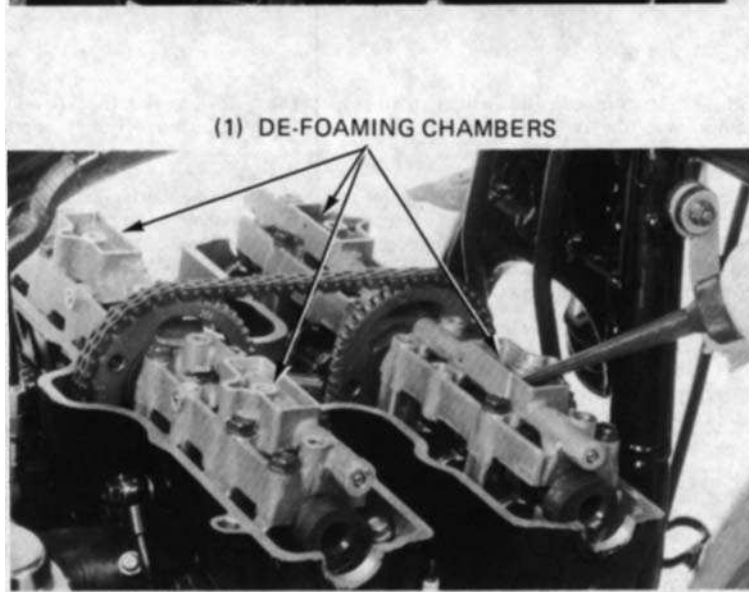
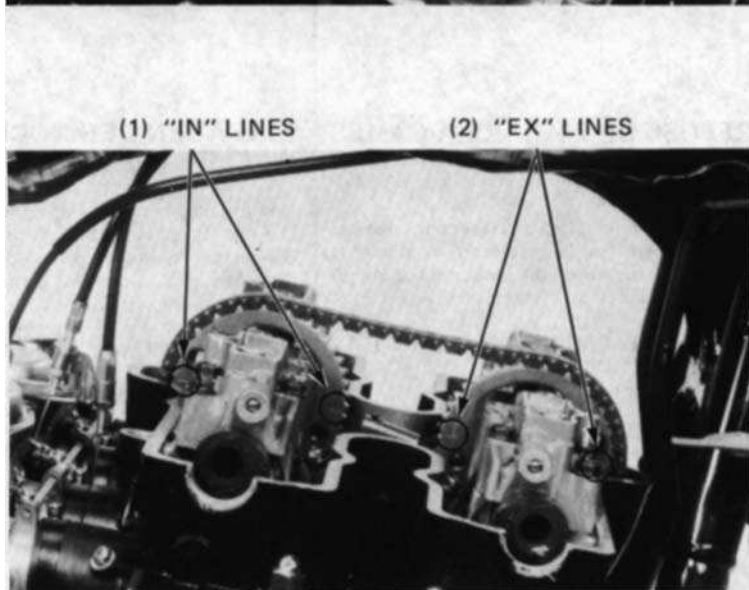
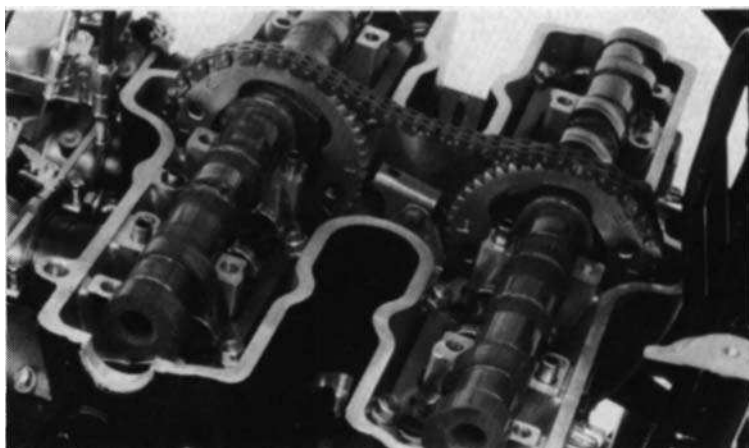
Place the cam chain on the sprockets.

Slide the cam sprockets onto the camshaft flanges.

Fill the de-foaming chambers with clean engine oil.

CAUTION;

Do not turn the camshaft before filling de-foaming chambers





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6. Cylinder Head & Valves

Align the cam sprocket bolt holes by turning the crankshaft slightly
Install and tighten the cam sprocket bolts

TORQUES: 18-20 Nm
(1.8-2.0 kg-m, 13-14 ft lb)

Turn the crankshaft clockwise (viewed from the right side of the engines) and re-align the "T" mark on the pulse rotor with the index mark on the crankcase.
Make sure that the "IN" lines on the intake cam sprocket and the "EX" lines on the exhaust cam sprocket align with the top of the cylinder head.

Push the cam chain tensioner lock pin forward to release it from the lock plate.

Install the cam chain guide.
Install the oil pipes with the oil bolts and sealing washers.
Tighten the camshaft holder bolts and the oil bolts to the specified torques in criss-cross pattern in 2-3 steps.

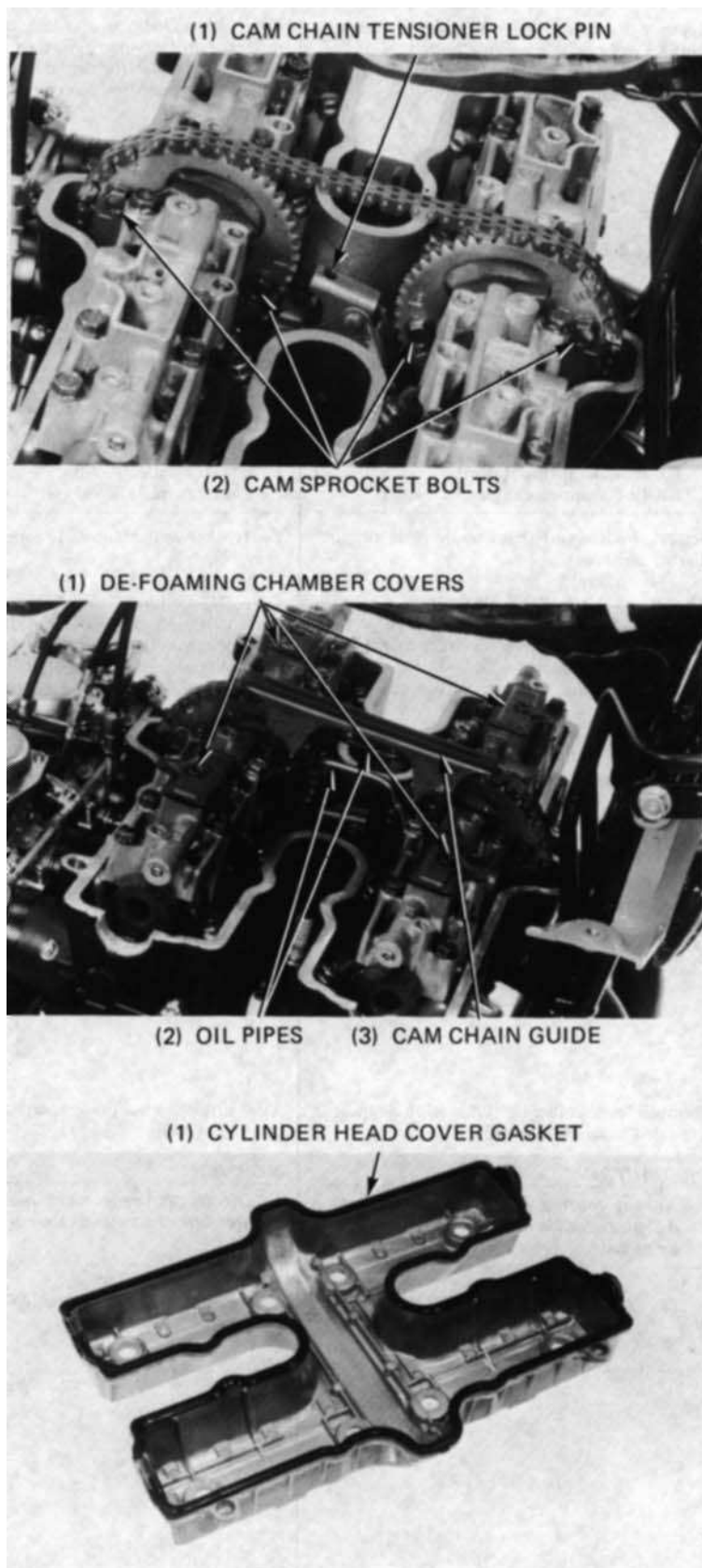
TORQUES:
CAMSHAFT HOLDER BOLTS:
12-16 Nm (1.2-1.6 kg-m, 9-12 ft.lb)
OIL BOLTS:
10-14 Nm (1.0-1.4 kg-m, 7-10 ft.lb)

Install the de-foaming chamber covers with the socket bolts.

Check the cylinder head cover gasket for damage or deterioration. Replace it if necessary.

NOTE:
To hold the gasket on the cylinder head cover, slightly apply adhesive agent to the several points on the cover.

Apply liquid sealant to the semicircular portions of the cylinder head.





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6. Cylinder Head & Valves

Install the cylinder head cover and tightens the cylinder head cover bolts.

NOTE

The two front bolts are the locating bolts. Tighten those bolts first.

Install the ignition coil, fuel tank and fairing.

